

KSA Office IT Support & Infrastructure Management

Phase 03: Cisco Packet Tracer Network Implementation

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Module 01: Network Topology Design

The implementation began by creating a new Cisco Packet Tracer project and designing the complete enterprise topology. The network consisted of one Cisco 2911 router, one core switch, two access switches, servers, network storage, printers, a wireless access point, and representative client devices from every department. Devices were named according to enterprise standards and positioned to represent a realistic office environment.

Module 02: Physical Cabling

All network devices were interconnected using the appropriate copper cabling. Infrastructure devices, servers, printers, wireless access points, and client computers were connected to the correct switches while trunk and access connections were planned according to enterprise design principles.

Module 03: Core Switch Configuration

The core switch was configured with a hostname, encrypted passwords, console and remote access security, password encryption, login banner, and saved configuration. This established the management and security baseline for the network.

Module 04: Access Switch Configuration

Both access switches were configured using the same standardized security configuration to ensure consistency throughout the network infrastructure.

Module 05: VLAN & Trunk Configuration

Departmental VLANs were created for Management, HR, Finance, Sales, Customer Support, Operations, IT, Reception, Warehouse, Servers, Printers, and Wireless. Access ports were assigned to their respective VLANs, and trunk links were configured between the core switch, access switches, and router.

Module 06: Inter-VLAN Routing

Router-on-a-Stick was implemented by configuring 802.1Q sub interfaces for every VLAN. Each VLAN received its own gateway, allowing secure communication between departments while maintaining logical separation.

Module 07: DHCP Services

DHCP pools were configured for every VLAN with reserved infrastructure addresses. Client devices automatically received IP addresses, subnet masks, gateways, DNS settings, and domain information. Configuration issues were diagnosed and corrected through systematic troubleshooting.

Module 08: Network Validation

Connectivity testing confirmed successful communication between VLANs, infrastructure devices, servers, printers, and representative workstations. DHCP operation and routing functionality were validated before proceeding to network hardening.

Module 09: Network Security & Hardening

Enterprise security practices were implemented by enabling SSH Version 2, creating a privileged administrator account, disabling Telnet, generating RSA keys, configuring Port Security with sticky MAC addresses, enabling PortFast and BPDU Guard on user-facing ports, and disabling unused switch interfaces.

Module 10: Final Validation & Documentation

The completed implementation was verified using Cisco IOS validation commands to confirm VLANs, trunk links, routing, DHCP, SSH, spanning tree, and port security. Final connectivity tests were completed, configurations were saved, and screenshots were collected to support the project documentation.

Project Outcome

This phase demonstrates the complete deployment of a secure enterprise network from planning through implementation, testing, troubleshooting, security hardening, and validation. The project shows practical Cisco networking skills relevant to IT Support, Network Support, System Administration, and Infrastructure roles.